Dominion Energy Virginia reliability: what public data shows, and where I'd look first

Independent analysis of public data

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3 November 2026

Data: EIA-861 · PUDL · NOAA Storm Events · EIA-930

Outage minutes rose from 130 to 133 because interruptions became more frequent, not longer

130 → 133

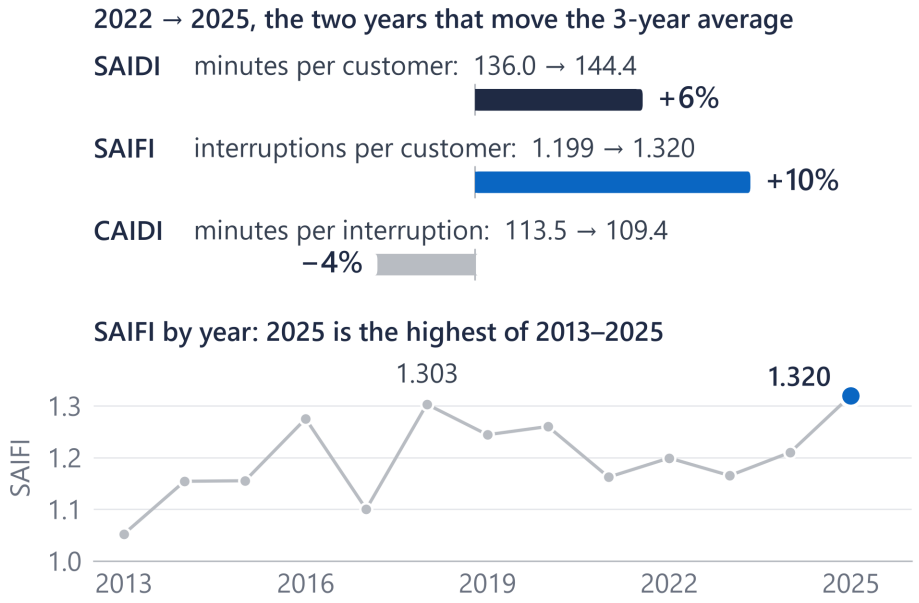
Reported 3-year average outage minutes per customer, ex-major events (2022–24 → 2023–25)

130.2 → 133.0

The same figures rebuilt from public EIA-861 data: they reconcile

136.0 → 144.4

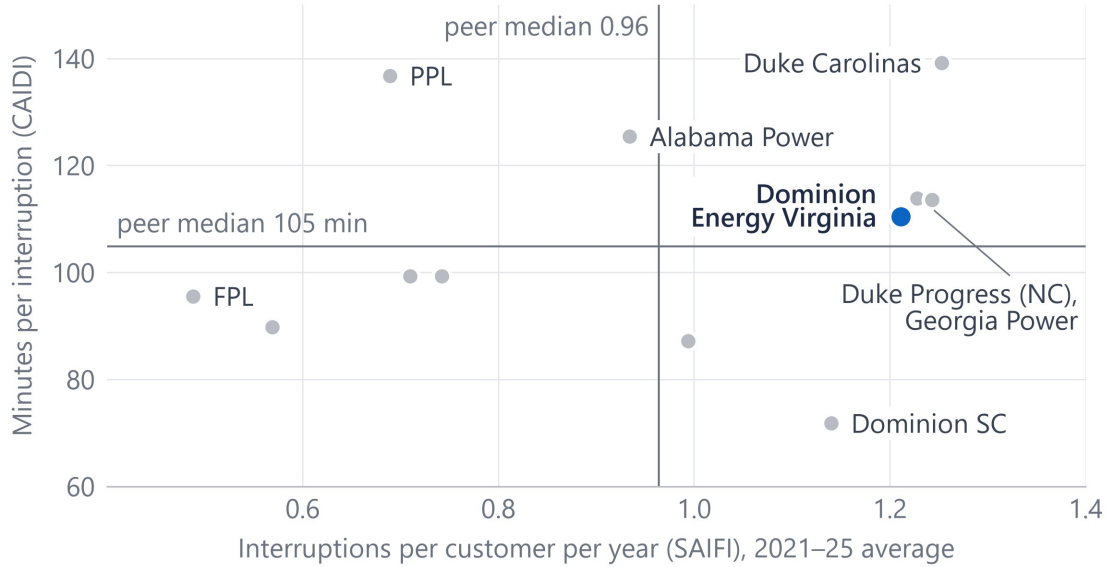
The average moved because 2025 replaced 2022 in the 3-year window



For Dominion: the first question is what is making interruptions more frequent, before how to restore faster.

Source: EIA-861 2025 final and PUDL (EIA-861) 2013–2024, Dominion Energy Virginia, excluding major event days; 3-year means of annual values. Reported figures: Dominion Energy 2025 Annual Report, p. 10.

Among 12 peer utilities, Dominion restores at a mid-pack pace; frequency is the gap



1.21

interruptions per customer a year: 4th highest of 12 (peer median 0.96)

110 min

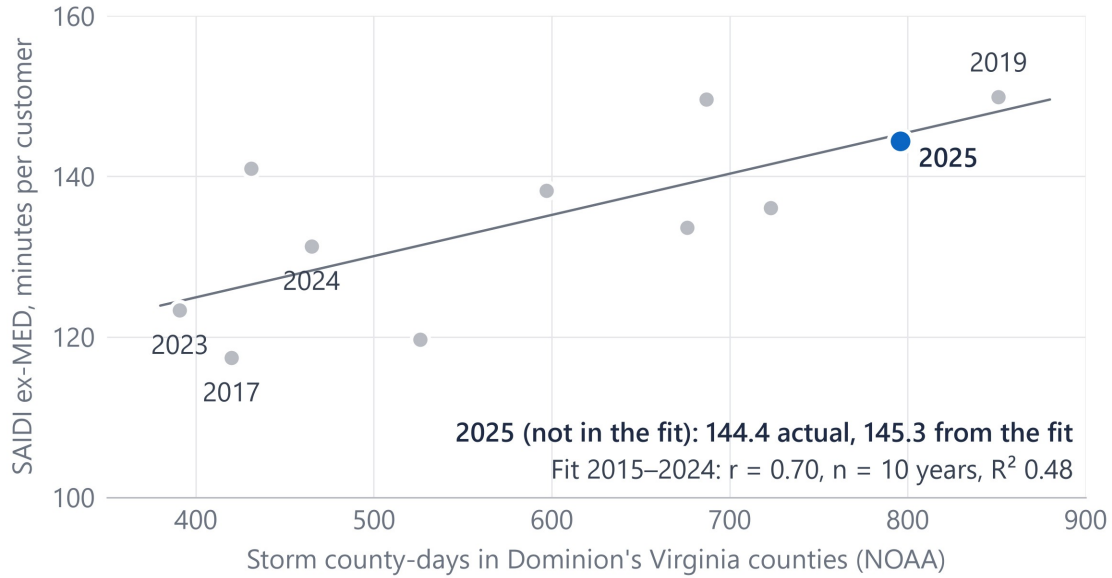
per interruption: 6th of 12 (peer median 105)

Methods differ: Dominion uses its own major-event rule, most peers IEEE 1366, so ranks are indicative.

For Dominion: preventing interruptions is where the gap to peers is; restoration speed is already near the middle.

Source: EIA-861 / PUDL 2021–2025, excluding major event days, means of annual values; 12 large investor-owned utilities in PJM and the Southeast, chosen by EIA ID before results.

Years with more storm days had more outage minutes (r = 0.70 over ten years)



465 → 796

storm county-days in Dominion's Virginia counties, 2024 → 2025

r = 0.70

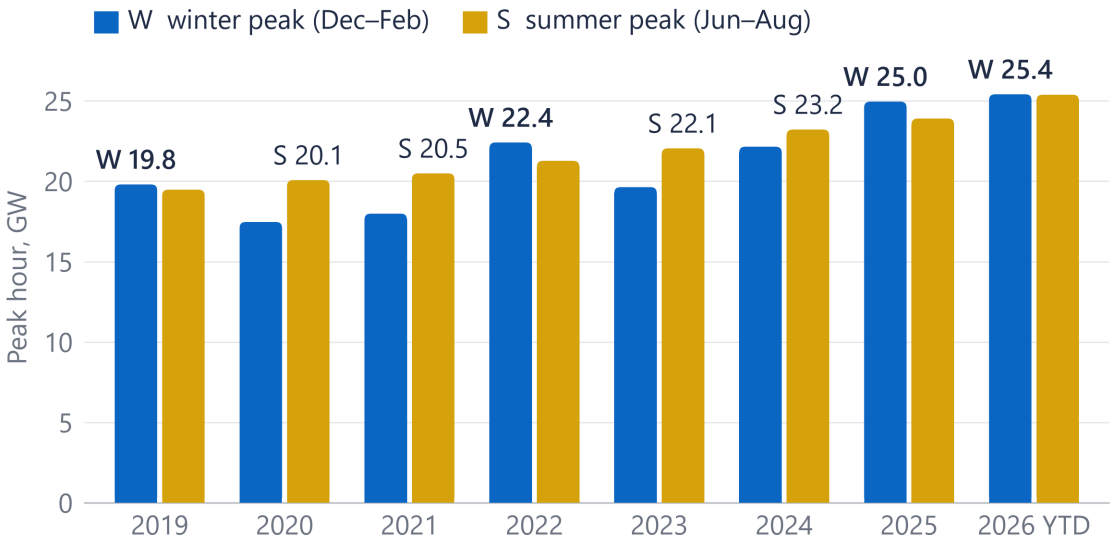
over 2015–2024 (n = 10): an association, not a cause

Dominion attributes the rise primarily to increased storm activity (2025 Annual Report, p. 10).

For Dominion: report reliability next to weather exposure, so that a stormy year is not mistaken for a trend.

Source: NOAA Storm Events 2015–2025, wind, winter and tropical event types, by county in Dominion's Virginia service territory (EIA-861); EIA-861 SAIDI excluding major event days; least-squares fit on 2015–2024.

Overnight demand grew faster than afternoon demand, and winter rivals summer for the yearly peak



+39–41%

average load in the hours ending 1–5 a.m., 2019 → 2025

about +27%

in the hours ending 1–5 p.m., same years

Lowest hour of the year +52%, highest +26%. 2026 so far: 25,413 MW winter vs 25,384 MW summer.

For Dominion: less overnight slack and winter peaks: plans and reports built around summer afternoons need updating.

Source: EIA-930 hourly demand, PJM Dominion (DOM) zone, which is wider than Dominion's retail territory; July 2018–October 2026 (2026 year to date). Winter = Dec–Feb, summer = Jun–Aug.

With Dominion's outage, meter and circuit data, I would start where public data stops

Days 1–30

Reconcile

Learn how outages, meter events and circuits are recorded and reported.

Rebuild one year's SAIDI, SAIFI and CAIDI and reconcile them to the filed figures.

Deliverable: a reconciled baseline with written definitions

Days 31–60

Diagnose

Interruptions by circuit and cause code, vegetation included; repeat interruptions (CEMI).

Restoration timed from meter “last gasp” (99.8% of meters are smart).

Deliverable: tested views and a data dictionary

Days 61–90

Prioritise and track

Rank feeders by customer-minutes; track Grid Resiliency work such as mainfeeder hardening against similar untreated feeders and weather.

Deliverable: a Power BI programme-tracking report, refreshed on a schedule

For Dominion: each step turns a system-level trend into a circuit-level view the programme team can act on.

Source: A plan: Dominion data is named by type only, and none was used here. Smart meters: EIA-861 2025 (AMI share of meters). Grid Resiliency: Dominion Energy website, read 1 November 2026.

Built like governed production reporting: tested, documented, and reconciled to the reported numbers



12 tests pass

keys, SAIDI = SAIFI × CAIDI, EIA's published 2025 values

Data dictionary

a business definition for every column and measure

Decisions log

14 decisions, such as storm types and peers, made before results

Reconciled

rebuilt 130.2 and 133.0 vs 130 and 133 reported

For Dominion: reconcile first, test every load, document every decision: the same discipline for Dominion's reporting.

Source: Project build: SQL scripts, tests T1–T12, data dictionary and decisions log D1–D14; row counts as loaded.

Three takeaways: frequency first, reliability net of weather, plan for a new load shape

1 **Look at frequency first**

Interruptions per customer +10% (2022→2025), the 2013–2025 high; 4th of 12 peers

2 **Net out the weather**

Storm days track outage minutes (r = 0.70, ten years): an association, not a cause

3 **Plan for a new load shape**

Hours ending 1–5 a.m. +39–41% (2019→2025); winter set the peak in 2019, 2022 and 2025

I build governed Snowflake → Power BI reporting, and I've already started learning your reliability data.

How my experience maps Mercury Insurance, Data Analyst

Snowflake pipeline, 20M+ rows	**Root-cause analysis on joined data**	**Validation and reconciliation**	**Metadata and business definitions**	**Requirements with operations and IT**
→ this mart	→ SAIFI by cause	→ reconciled 133.0	→ data dictionary	→ programme teams

For Dominion: I would like to do this with your data.

Source: Numbers from slides 2–5. Experience: Mercury Insurance, Data Analyst, 2026 to present.

Definitions: SAIDI = SAIFI × CAIDI, and major event days are set aside by rule

SAIDI
outage minutes per customer per year

=

SAIFI
interruptions per customer per year: how often

×

CAIDI
minutes per interruption: how long

Dominion 2025: 1.320 × 109.4 = 144.4 minutes (excluding major event days)

Major event days (MED) are set aside so that one extreme storm does not swamp the trend. IEEE 1366 marks a day as a major event day when its SAIDI exceeds a threshold set from the previous five years of daily SAIDI (the “2.5 beta” method).

Dominion reports to EIA under its own major-event rule; most peers in this deck use IEEE 1366. Headline figures here exclude major event days; all-event figures are on A3.

Log split: ln(SAIDI ratio) = ln(SAIFI ratio) + ln(CAIDI ratio). Share from frequency = ln(SAIFI ratio) ÷ ln(SAIDI ratio): 2023→2025, 0.125 ÷ 0.158 = 79%.

Source: IEEE Std 1366 (reliability indices); EIA-861 reliability, Dominion Energy Virginia 2023 and 2025.

The weather fit is a check on one year, not a forecast

Year	Storm county-days	SAIDI ex-MED	From the fit	Difference
2015	526	119.6	131.4	−11.8
2016	597	138.2	135.1	+3.2
2017	420	117.4	126.0	−8.5
2018	687	149.6	139.7	+9.9
2019	851	149.9	148.1	+1.8
2020	431	141.0	126.5	+14.5
2021	676	133.6	139.1	−5.5
2022	723	136.0	141.5	−5.5
2023	391	123.3	124.5	−1.1
2024	465	131.3	128.3	+3.0
2025 (not in fit)	**796**	**144.4**	**145.3**	**−0.9**

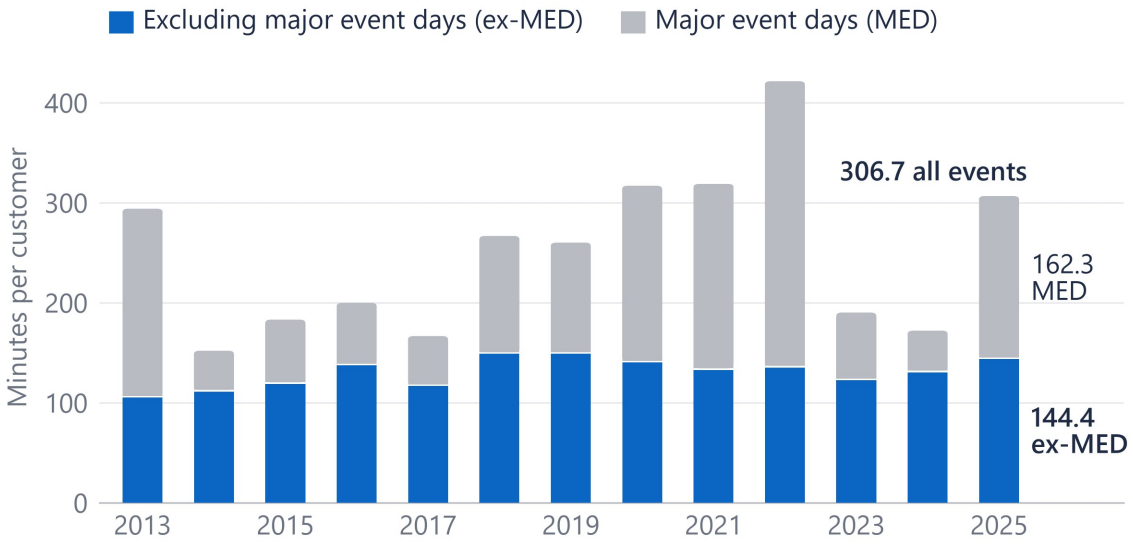
Fit on 2015–2024 (n = 10): SAIDI = 104.4 + 0.0514 × storm county-days; r = 0.70, R^2 0.48.

2025 was left out: the fit gives 145.3, the actual was 144.4.

Adjusted for storm days, no clear trend since 2015: about +0.3 minutes a year (95% range −1.4 to +2.1).

Source: NOAA Storm Events and EIA-861, 2015–2025; ordinary least squares on ten annual points (exploratory).

Major event days were about half of all-event outage minutes in 2025



53%

of 2025's all-event outage minutes fell on major event days (162.3 of 306.7)

Major-event minutes swing most: 285 in 2022, 41 in 2024.

Major-event minutes are EIA's all-events SAIDI minus ex-MED SAIDI, not a storm attribution.

Source: EIA-861 / PUDL 2013–2025, Dominion Energy Virginia: SAIDI with and without major event days.

Every number traces to a public source and a tested query

Source	Publisher	Years	Grain	Rows
EIA-861 Reliability, 2025 final	EIA	2025	utility × state × year	977
PUDL EIA-861 reliability	Catalyst Cooperative	2013–2024	utility × state × year	24,358
EIA-861 Service Territory	EIA	2025	utility × county	11,803
NOAA Storm Events	NOAA NCEI	2013–2025	storm event	831,288
EIA-930 hourly demand	EIA	Jul 2018–Oct 2026	region × hour	6,077,232
Dominion Energy 2025 Annual Report	Dominion Energy	2025	document (130, 133)	reference

Tests after each load (T1–T12, all pass): no duplicate keys; SAIDI = SAIFI × CAIDI within 2%; Dominion 2025 equals EIA's published 306.7, 144.4 and 2,675,735 customers; every county mapped; full years; no duplicate hours.

Source: Project documentation: sources register (publisher, vintage, licence) and data-quality tests; all public data.

What public data cannot show, and what would answer it

Public data limit	What would answer it (Dominion data, by type)
One number per utility per year (EIA-861)	Outage records by event, circuit and cause
Storms counted by county, not by line	Outage records with cause codes and circuit location
Methods differ between utilities	Dominion's daily SAIDI and its filing method
The DOM zone is wider than the retail territory	Meter interval data by circuit
No customer identities: growth can't be attributed	Customer class and large-load records

Source: Project scope limits (decision D14) and the sources above.

The twelve-peer comparison, with each utility's reporting method

Utility	Major-event method	SAIFI 2021–25	CAIDI 2021–25	SAIDI 2023–25
Duke Carolinas (NC)	IEEE 1366	1.25	139	179
Georgia Power	Mixed	1.24	114	141
Duke Progress (NC)	IEEE 1366	1.23	114	144
Dominion Energy Virginia	**Own method**	**1.21**	**110**	**133**
Dominion Energy SC	IEEE 1366	1.14	72	84
Duke Energy Florida	Mixed	0.99	87	83
Alabama Power	Own method	0.93	125	110
PECO	IEEE 1366	0.74	99	66
BGE	IEEE 1366	0.71	99	64
PPL Electric	IEEE 1366	0.69	137	99
Pepco (MD)	IEEE 1366	0.57	90	48
FPL	IEEE 1366	0.49	96	47

Source: EIA-861 / PUDL, excluding major event days; means of annual values; peers chosen by EIA ID before results. Sorted by SAIFI.